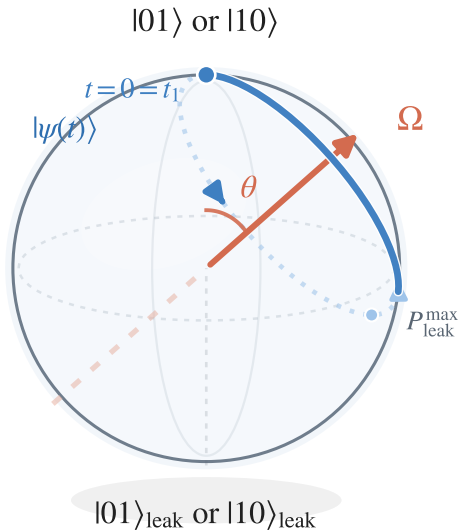


(b)



$$\Omega = (J_{\text{int}}, 0, \Delta)$$

$$\tan \theta = J_{\text{int}} / \Delta$$

$$\Omega = \sqrt{\Delta^2 + J_{\text{int}}^2}$$

$$t_n = 2\pi n / \Omega$$

closed loop $\Rightarrow$ zero residual leakage